



Refresher Course

90 Practice Questions and Answers for STCW Exams

Updated for DG Shipping India Exams | Source: brightmariner.com

Practice Set 1

Q1. Fire Triangle Fill up the blanks

- A) Fuel
- B) Heat
- C) Oxygen
- D) All of the above

Correct Answer: D (All of the above)

Explanation: The fire triangle consists of three essential elements: fuel, heat, and oxygen. Removing any one of these elements will result in the extinguishment of the fire.

Q2. Class A involves flammable liquids. True or False?

- A) True
- B) False

Correct Answer: B (False)

Explanation: Class A fires involve solid combustible materials like wood, paper, and textiles. Flammable liquids fall under Class B, making the statement false.

Q3. What is the minimum capacity of emergency fire pumps in meter cube per hour?

- A) 15 meter cube per hour
- B) 20 meter cube per hour
- C) 25 meter cube per hour
- D) 30 meter cube per hour

Correct Answer: C (25 meter cube per hour)

Explanation: Under SOLAS requirements for fire safety systems, the emergency fire pump must have a minimum capacity to deliver a specific volume of water, which is established as 25 cubic meters per hour for cargo ships.

Q4. What is the best fire extinguisher for electrical fires?

- A) CO2
- B) Foam
- C) Water
- D) DCP

Correct Answer: A (CO2)

Explanation: CO2 is the preferred medium for electrical fires because it is non-conductive, leaves no residue, and effectively displaces oxygen without damaging sensitive electrical components.



- A) Green
- B) Yellow
- C) White
- D) Blue

Correct Answer: C (White)

Explanation: According to the COLREGs, white light is required for the all-round masthead and stern lights on smaller vessels to ensure visibility in all directions, acting as a navigational signal.

Q6. Fire extinguishing system required for propelling machines and diesel-driven motors:

- A) Hyper mist
- B) CO2
- C) DCP
- D) Foam

Correct Answer: A (Hyper mist)

Explanation: Hyper mist (Water mist) systems are highly effective for engine room machinery as they cool the fire and provide localized protection without the oxygen-depleting risks of CO2.

Q7. How many hours should a diesel-driven fire pump run?

- A) 12 hours
- B) 14 hours
- C) 16 hours
- D) 18 hours

Correct Answer: D (18 hours)

Explanation: Under SOLAS requirements, the emergency fire pump must be capable of continuous operation for at least 18 hours to ensure sufficient firefighting capability during a prolonged emergency.

Q8. Dry chemical powder extinguisher works on what principle?

- A) Smothering
- B) Starvation
- C) Inhibition
- D) Cooling

Correct Answer: C (Inhibition)

Explanation: Dry Chemical Powder (DCP) extinguishes fire primarily by chemical inhibition, where the powder interferes with the free radicals in the combustion chain reaction.

Q9. Class B bulkheads are classified as B15 and B0. True or False?

- A) True
- B) False

Correct Answer: A (True)

Explanation: Class B divisions are structural fire protection barriers; B15 prevents smoke/flame and limits temperature rise for 15 minutes, while B0 prevents smoke/flame but lacks insulation requirements.



- A) 300 degrees centigrade
- B) 350 degrees centigrade
- C) 400 degrees centigrade
- D) 250 degrees centigrade

Correct Answer: D (250 degrees centigrade)

Explanation: According to the FTP Code, the average temperature on the unexposed side of a B class division must not rise more than 225 degreesC above initial, with a maximum of 250 degreesC at any point.

Q11. DCP firefighting system is used in which types of vessels?

- A) Oil tankers
- B) Chemical tankers
- C) Bulk carriers
- D) Gas carriers

Correct Answer: D (Gas carriers)

Explanation: DCP systems are primarily used on gas carriers because the powder is chemically inert and effectively tackles fires involving liquefied gases where other agents may be ineffective.

Q12. What is the maximum weight of a portable extinguisher?

- A) 16 kgs
- B) 25 kgs
- C) 12 kgs
- D) 23 kgs

Correct Answer: D (23 kgs)

Explanation: Portable fire extinguishers under SOLAS/FSS Code regulations are limited to a maximum mass of 23 kg to ensure they are manageable by a single crew member during an emergency.

Q13. Class A fire extinguisher is best used on:

- A) Solids
- B) Liquids
- C) Gases
- D) Metals

Correct Answer: A (Solids)

Explanation: Class A fires involve ordinary combustible solids like wood, paper, and textiles. Water is the primary extinguishing agent as it provides a cooling effect.

Q14. In which language is the fire control plan written on board?

- A) French
- B) English
- C) Spanish
- D) Any local language

Correct Answer: B (English)

Explanation: As per SOLAS requirements, the fire control plan must be written in the working language of the ship and in either English, French, or Spanish.



- A) Accommodation
- B) Main deck
- C) Engine room
- D) None of the above

Correct Answer: D (None of the above)

Explanation: Automatic CO2 systems are not used in occupied spaces like the accommodation or engine room due to the extreme danger of suffocation to crew members.

Q16. How often should rescue boards be launched with their assigned crew and manning water?

- A) Every month
- B) Every week
- C) Once in three months
- D) Once in two weeks

Correct Answer: C (Once in three months)

Explanation: SOLAS regulations require that rescue boats be launched with their assigned crew, and the boat must be maneuvered in the water at least once every three months.

Q17. Your vessel is in collision and you are not sure if your vessel is sinking. How will you handle the situation?

- A) Search for missing persons, monitor the situation, prepare for abandonment if required.
- B) Leave the vessel and stay close by.
- C) Wait and see the situation.
- D) Ensure the lifeboat has enough fuel to reach land.

Correct Answer: A (Search for missing persons, monitor the situation, prepare for abandonment if required.)

Explanation: The primary response to a collision is to assess the damage and structural integrity. Monitoring the situation is crucial before committing to abandoning the vessel.

Q18. The visible range of parachute flares at nighttime is:

- A) 8 nautical miles
- B) 10 nautical miles
- C) 15 nautical miles
- D) 25 nautical miles

Correct Answer: D (25 nautical miles)

Explanation: According to LSA code, parachute flares are required to have a high intensity and visibility, capable of being seen at significant distances to alert SAR assets.

Q19. What should be done in case of a fire on board?

- A) Let the wind follow the fire to reduce oxygen.
- B) Let the stem be up against the wind.
- C) Continue on course and speed.
- D) Reduce speed, keep the fire zone leeward.

Correct Answer: D (Reduce speed, keep the fire zone leeward.)

Explanation: Maneuvering to keep the fire leeward ensures that the wind blows smoke and flames away from the rest of the ship, assisting the fire-fighting team.



- A) Pull, Aim, Shut, Sit
- B) Power, Pull, Aim, Squeeze, Sweep
- C) Pass, Aim, Sure, Speed
- D) Pull, Aim, Squeeze, Sweep

Correct Answer: D (Pull, Aim, Squeeze, Sweep)

Explanation: PASS is the standard mnemonic for operating portable fire extinguishers: Pull the pin, Aim at the base, Squeeze the handle, and Sweep from side to side.

Q21. How should the painter of a life raft, which is fitted with an HRU, be secured to the ship?

- A) Secured via a weak link to a secure point.
- B) Secured to a part of the hydrostatic release unit that is designed to break free.
- C) It should not be secured in any way.
- D) Directly to a secure point on the ship.

Correct Answer: A (Secured via a weak link to a secure point.)

Explanation: The painter is secured via a weak link so that if the vessel sinks, the raft will float free, and the link will break once the raft is inflated, preventing it from being dragged down.

Q22. Before lifting a lifeboat from water, it must be ensured that:

- A) The bottom plug is shipped.
- B) Fall wires are clear and looped securely.
- C) The engine is going astern.
- D) The anchor is streamed.

Correct Answer: B (Fall wires are clear and looped securely.)

Explanation: Ensuring the fall wires are clear and looped securely prevents fouling or jamming, which could lead to accidental or uneven release during the recovery of the lifeboat.

Q23. What is the duration of an emergency escape breathing device (EBD)?

- A) 5 minutes
- B) 8 minutes
- C) 12 minutes
- D) 10 minutes

Correct Answer: D (10 minutes)

Explanation: EEBDs are designed for personnel to escape a fire-affected space; SOLAS requires them to have a service duration of at least 10 minutes.

Q24. A self-contained air breathing apparatus bottle with 1200 liters of air will last for how many minutes?

- A) 20 minutes
- B) 25 minutes
- C) 30 minutes
- D) 45 minutes

Correct Answer: C (30 minutes)

Explanation: Standard breathing rate is roughly 40 liters/min. A 1200-liter bottle lasts for 30 minutes ($1200 / 40 = 30$), accounting for standard consumption during fire operations.

Q25. The extinguisher that can be used for class F or class K fires is:



- A) CO2
- B) Dry chemical powder
- C) Foam
- D) Wet chemical

Correct Answer: D (Wet chemical)

Explanation: Wet chemical extinguishers are specifically designed for Class F/K fires, typically involving cooking oils and fats, as they form a soap-like film to prevent re-ignition.

Q26. Oxygen removal stops fire by:

- A) Cooling effect
- B) Inhibition
- C) Starvation
- D) Smothering

Correct Answer: D (Smothering)

Explanation: Smothering involves depriving the fire of oxygen, one of the primary elements of the fire triangle, thereby extinguishing the combustion process.

Q27. What is the fourth component of a fire tetrahedron?

- A) Molecular chain reaction
- B) Inhibition
- C) Fire reaction
- D) None of the above

Correct Answer: A (Molecular chain reaction)

Explanation: The fire tetrahedron adds the 'Chemical Chain Reaction' to the classic fire triangle (Fuel, Heat, Oxygen), explaining why chemical inhibition is effective.

Q28. What is the color coding of a portable water extinguisher?

- A) Cream
- B) Yellow
- C) Blue
- D) Signal red

Correct Answer: D (Signal red)

Explanation: Portable fire extinguishers are color-coded for identification. Under international standards, water extinguishers are identified by a signal red body color.

Q29. Which type of extinguishers are difficult to recharge on board?

- A) CO2
- B) DCP
- C) Foam
- D) Water

Correct Answer: A (CO2)

Explanation: CO2 extinguishers require high-pressure recharging facilities and specialized equipment that are typically not maintained or available on board a standard merchant vessel.



- A) 10 meters
- B) 50 meters
- C) 20 meters
- D) 30 meters

Correct Answer: D (30 meters)

Explanation: SOLAS requires that the lifeline of a self-contained breathing apparatus must be fireproof and have a minimum length of 30 meters.

Practice Set 2

Q1. Which dangerous condition should be kept in mind after using a 15-pound CO2 fire extinguisher on a small oil fire on the engine room floor plates?

- A) Possible suffocation from the CO2 gas.
- B) Forgetting to secure the engine room ventilation system.
- C) CO2 gas being drawn into nearby engine or air compressor intakes.
- D) Rekindling of the fire.

Correct Answer: D (Rekindling of the fire.)

Explanation: CO2 is not a cooling agent. After using it, the danger remains that the hot fuel or surfaces may cause the fire to reignite once the gas dissipates.

Q2. Which of the listed classes of fire is considered as one involving electrical equipment?

- A) Class "A"
- B) Class "B"
- C) Class "C"
- D) Class "D"

Correct Answer: C (Class "C")

Explanation: Class C fires involve energized electrical equipment. Using conductive agents like water would be hazardous, necessitating non-conductive agents like CO2.

Q3. Which of the following would be considered a Class "B" fire?

- A) Electrical fire in a motor
- B) Waste rag fire in the engine room
- C) Oil fire in the engine room bilges
- D) Fire in the main switchboard

Correct Answer: C (Oil fire in the engine room bilges)

Explanation: Class B fires involve flammable liquids. Engine room bilges often contain oil, making them a prime location for Class B fire risks.



- A) Magnesium
- B) Paper
- C) Wood
- D) Diesel Oil

Correct Answer: D (Diesel Oil)

Explanation: Class B fires are specifically defined as fires involving flammable liquids, such as diesel oil, gasoline, or paints.

Q5. Which of the listed extinguishing agents is NOT suitable for fighting a liquid paint fire?

- A) Dry chemical
- B) Foam
- C) Water
- D) Carbon dioxide

Correct Answer: C (Water)

Explanation: Water should not be used on liquid paint fires as it can spread the burning liquid and increase the surface area of the fire.

Q6. Which extinguishing agent is most effective on a mattress fire?

- A) CO2
- B) Foam
- C) Dry Chemical
- D) Water

Correct Answer: D (Water)

Explanation: Water is the most effective agent for mattress fires because it penetrates deep into the material to cool the seat of the fire, preventing deep-seated re-ignition.

Q7. The extinguishing agent most effective for combating wood fires is:

- A) Water
- B) Carbon dioxide
- C) Foam
- D) Dry chemical

Correct Answer: A (Water)

Explanation: Water is ideal for wood fires (Class A) due to its high latent heat of vaporization, which effectively cools the material below its ignition temperature.

Q8. You are fighting a fire in a watertight compartment using hoses and river water. Stability may be reduced because of:

- A) Progressive down flooding
- B) Reduction of water in the storage tanks
- C) Increase in free surface which reduces the metacentric height
- D) Reduction of KG to the minimum allowable

Correct Answer: C (Increase in free surface which reduces the metacentric height)

Explanation: Free surface effect occurs when a compartment is partially filled with water, causing the center of gravity to shift and significantly reducing the metacentric height (GM), affecting stability.



- A) Is effective if used promptly on an oil fire
- B) Has a greater cooling effect than water
- C) Is lighter than air
- D) Is harmless to cargo and crew

Correct Answer: A (Is effective if used promptly on an oil fire)

Explanation: CO2 is a clean, non-conductive gas that effectively displaces oxygen, making it ideal for immediate response to oil fires in confined spaces.

Q10. Carbon dioxide as a firefighting agent has which advantage over other agents?

- A) It causes minimal damage.
- B) It is safer for personnel.
- C) It is cheaper.
- D) It is most effective on a per unit basis.

Correct Answer: A (It causes minimal damage.)

Explanation: CO2 is a non-residue-forming gas, meaning it causes minimal damage to sensitive machinery or electronic equipment compared to water or powder.

Q11. To prevent the spread of fire by convection you should:

- A) Cool the bulkhead around the fire
- B) Remove combustibles from direct exposure
- C) Close all openings to the area
- D) Shut off all electrical power

Correct Answer: C (Close all openings to the area)

Explanation: Convection spreads fire via the movement of hot gases. Closing all openings (like fire doors and dampers) restricts the flow of these gases, containing the fire.

Q12. What is NOT a characteristic of carbon dioxide fire-extinguishing agents?

- A) Effective even if ventilation is not shut down
- B) Will not deteriorate in storage
- C) Non-corrosive
- D) Effective on electrical equipment

Correct Answer: A (Effective even if ventilation is not shut down)

Explanation: CO2 is effective only if the space is sealed. If ventilation is not shut down, the gas will dissipate, and oxygen will return, allowing the fire to continue.

Q13. Foam extinguishes a fire by:

- A) Destroying the burning materials
- B) Chemical reaction with the burning material
- C) Absorbing the burning material
- D) Smothering the burning material

Correct Answer: D (Smothering the burning material)

Explanation: Foam works by creating a barrier (blanket) over the liquid fuel, which cuts off the oxygen supply (smothering) and prevents the release of flammable vapors.



- A) Carefully apply a low velocity fog over the foam blanket
- B) Wade through the foam blanket to evenly distribute the foam
- C) Be careful not to disturb the foam blanket
- D) Sweep the foam blanket overboard and secure firefighting equipment

Correct Answer: C (Be careful not to disturb the foam blanket)

Explanation: Disturbing the foam blanket allows oxygen to reach the fuel, potentially causing the fire to reignite. The blanket must remain intact until the area is confirmed safe.

Q15. The most likely location for a liquid cargo fire to occur on a tanker would be:

- A) In the midships house
- B) At the main deck manifold
- C) At the vent header
- D) In the pump room

Correct Answer: D (In the pump room)

Explanation: The pump room is a high-risk area on tankers due to the concentration of piping, valves, and moving parts associated with cargo transfer, often carrying flammable liquids.

Q16. You should be most concerned about a possible explosion or fire in fuel tanks:

- A) During fueling when the fuel first strikes the tank bottom
- B) During fueling when the fuel strikes fuel already in the tank
- C) When underway as the fuel is moved by wave action
- D) Shortly after fueling when fuel vapors gather

Correct Answer: A (During fueling when the fuel first strikes the tank bottom)

Explanation: Fueling operations pose the highest risk of static electricity discharge or spill ignition when the initial stream strikes the tank bottom, creating turbulence and vapor.

Q17. You have carbon tetrachloride as part of the cargo. If a fire breaks out in the general area, what is the major danger from the carbon tetrachloride?

- A) It will explode if exposed to a flame
- B) Phosgene gas may be formed if it comes in contact with hot metal
- C) It will burn rapidly once ignited
- D) You cannot use water to fight the fire because it will react with the carbon tetrachloride

Correct Answer: B (Phosgene gas may be formed if it comes in contact with hot metal)

Explanation: Carbon tetrachloride decomposes when exposed to high heat or flame, producing phosgene gas, which is extremely toxic and lethal to personnel.

Q18. Radiation can cause a fire to spread by:

- A) Transmitting the heat of a fire through the ship's metal
- B) Burning liquids flowing into another space
- C) Heated gases flowing through ventilation systems
- D) The transfer of heat across an unobstructed space

Correct Answer: D (The transfer of heat across an unobstructed space)

Explanation: Radiation is the transfer of heat through electromagnetic waves across space, meaning it does not require direct contact or a medium to ignite nearby combustibles.



- A) Cool the bulkheads around the fire
- B) Remove combustibles from direct exposure
- C) Close all openings to the area
- D) Shut off all electric power

Correct Answer: A (Cool the bulkheads around the fire)

Explanation: Conduction transfers heat through solid materials like steel bulkheads. Cooling these bulkheads with water absorbs the heat and prevents the fire from reaching adjacent compartments.

Q20. You notice smoke coming from an open laundry room doorway. After activating the fire alarm, which of the following would you do FIRST?

- A) Attempt to determine what is burning.
- B) Acquire the nearest self-contained breathing apparatus.
- C) Break out the nearest fire hose.
- D) Wait for the fire team to arrive and assist as directed.

Correct Answer: A (Attempt to determine what is burning.)

Explanation: Immediate assessment is required to determine the nature and extent of the fire so that the correct firefighting strategy and agents can be utilized efficiently.

Q21. You are on watch at night in port and discover a fire in #1 hatch. Which action should you take FIRST?

- A) Advise the Chief Mate and Master
- B) Release carbon dioxide into the hatch
- C) Sound the general alarm
- D) Lead a fire hose to the hatch

Correct Answer: C (Sound the general alarm)

Explanation: Sounding the general alarm is the highest priority to alert the entire crew, initiate the mustering process, and organize the response team.

Q22. The spreading of fire as a result of heat being carried through a vessel's ventilation system is an example of heat transfer by:

- A) Conduction
- B) Convection
- C) Radiation

Correct Answer: B (Convection)

Explanation: Convection involves the movement of heated fluids (including air or gases). Ventilation systems act as channels for these hot gases to transfer heat to other parts of the ship.

Q23. Which of the following is NOT identified on the vessel's fire control plan?

- A) Gas detector
- B) Fire control plan
- C) Fire and emergency signals
- D) Dry chemical monitor

Correct Answer: C (Fire and emergency signals)

Explanation: The fire control plan shows the location of safety equipment; however, general emergency signals are specified in the Muster List and Safety Training Manual, not the fire control plan.



- A) Stowed separately in a clip or rack at each fire station
- B) Attached to the all-purpose nozzle by a chain
- C) Coated with heavy grease to prevent corrosion
- D) Painted red to be identified as emergency equipment

Correct Answer: B (Attached to the all-purpose nozzle by a chain)

Explanation: The high-velocity fog tip is attached by a chain to ensure it remains readily available at the fire station for immediate changeover, preventing loss during the urgency of firefighting.

Q25. The most common and widely used cooling agent used for fighting fires on merchant vessels is:

- A) Carbon dioxide
- B) Flue gas
- C) Water
- D) Steam smothering

Correct Answer: C (Water)

Explanation: Water is the most common and versatile cooling agent on ships. It effectively absorbs heat by changing from liquid to vapor, cooling burning surfaces, and creating a steam blanket to smother the fire.

Q26. The outlet at a fire hydrant may be positioned anywhere from horizontal to pointing:

- A) 45 degrees upward
- B) Vertically upward
- C) Vertically downward

Correct Answer: C (Vertically downward)

Explanation: Fire hydrant valves are often positioned pointing vertically downward. This orientation minimizes stress on the fire hose coupling and prevents sharp bends or kinks in the hose which could restrict water flow.

Q27. The primary function(s) of an automatic sprinkler system is(are) to:

- A) Extinguish the fire which triggers it
- B) Limit the spread of fire and control the amount of heat produced
- C) Protect people in the areas which have sprinkler heads
- D) Alert the crew to the fire

Correct Answer: B (Limit the spread of fire and control the amount of heat produced)

Explanation: Automatic sprinkler systems are primarily designed for fire control. They limit the spread of fire by cooling the area and surrounding structure while keeping the fire intensity manageable until active firefighting intervention arrives.

Q28. If a portable carbon dioxide fire extinguisher has been partially discharged, it should be immediately:

- A) Tagged and restowed properly in its rack
- B) Weighed and the charge loss recorded before being returned to its rack
- C) Given to the chief engineer for recharging
- D) Tagged 'empty' and stowed separately from charged extinguishers

Correct Answer: D (Tagged 'empty' and stowed separately from charged extinguishers)

Explanation: Any partially discharged extinguisher is unreliable and may not have sufficient pressure or agent for a real emergency. It must be tagged 'empty' and moved to a separate storage area to prevent it from being mistaken for a fully charged unit.



- A) Foam conducts electricity.
- B) To be most effective, foam should be directed at the base of the fire.
- C) Foam is most effective on burning liquids which are flowing.
- D) Foam can ONLY be used to extinguish class A fires.

Correct Answer: C (Foam is most effective on burning liquids which are flowing.)

Explanation: Foam creates a seal over the surface of burning liquids, making it highly effective for running fuel fires. It separates the fuel from oxygen, preventing reignition and effectively suppressing vapors.

Q30. Which of the following hazards is associated with fuel oil fires?

- A) Emission of noxious gases
- B) Reduced fire-fighting effectiveness due to low oxygen content
- C) Emission of highly flammable vapors
- D) Flashover of confined vapor clouds

Correct Answer: D (Flashover of confined vapor clouds)

Explanation: Fuel oil fires in confined engine room spaces can lead to rapid heat accumulation and the build-up of flammable vapors. If these vapors reach their lower explosive limit, a 'flashover' or explosion can occur, posing a severe threat to the vessel.

Practice Set 3

Q1. A carbon dioxide fire extinguisher is required to be recharged if the weight loss exceeds what percentage of the weight of the charge?

- A) One percent
- B) Five percent
- C) Seven percent
- D) Ten percent

Correct Answer: D (Ten percent)

Explanation: According to maritime regulations and manufacturers' standards, a carbon dioxide extinguisher must be recharged if the weight of the agent lost exceeds 10% of the original charged weight. This ensures the unit remains effective for the intended discharge duration.

Q2. Fire hydrant valves on a crude oil tanker shall be isolated from the crude oil washing system by

- A) Spade blanks
- B) Face blanks
- C) Non-return valves
- D) Automatic closing valves

Correct Answer: A (Spade blanks)

Explanation: Spade blanks (also known as slip blinds) provide a positive, physical barrier between the fire main and the crude oil washing system. This is a critical safety requirement to prevent hazardous hydrocarbon vapors from entering the fire main.



- A) Point the horn down
- B) Turn the cylinder upside-down
- C) Break the rupture disc
- D) Pull pin, squeeze grip

Correct Answer: D (Pull pin, squeeze grip)

Explanation: The standard procedure for operating a CO2 extinguisher is 'PASS': Pull the safety pin, Aim the nozzle at the base of the fire, Squeeze the operating handle, and Sweep from side to side.

Q4. All portable fire extinguishers must be capable of being

- A) Carried by hand to a fire
- B) Carried or rolled to a fire
- C) Recharged in the field
- D) Used on class HB" fires

Correct Answer: A (Carried by hand to a fire)

Explanation: Portable fire extinguishers are designed to be mobile so they can be immediately brought to the site of an incipient fire. Their size and weight are constrained by regulations to ensure a single crew member can transport and operate them.

Q5. Recharging a previously used cartridge-operated dry chemical fire extinguisher is accomplished by

- A) Authorized fire equipment servicing personnel only
- B) Replacing the propellant cartridge and refilling it with powder
- C) Puncturing the cartridge seal after installation
- D) Recharging the cartridge and refilling it with powder

Correct Answer: B (Replacing the propellant cartridge and refilling it with powder)

Explanation: For cartridge-operated extinguishers, the propellant charge (cartridge) and the fire-extinguishing powder are stored separately. After discharge, the cartridge must be replaced and the vessel must be refilled with the specified amount of powder to ensure it functions correctly.

Q6. Multiple fire pumps may be used for other purposes provided that one pump is

- A) On line to the fire main
- B) Kept available for use on the fire main at all times
- C) Capable of being connected to the fire main
- D) Rated at or above 125 psi

Correct Answer: B (Kept available for use on the fire main at all times)

Explanation: While fire pumps can be used for other essential duties like ballast or general service, at least one pump must always remain on standby or be immediately connectable to the fire main to ensure instantaneous readiness during an emergency.

Q7. Fire hoses should always be stowed at the fire main in a manner which will allow

- A) All sections to be quickly connected
- B) The nozzle end to be run out to the fire
- C) All sections to be quickly drained
- D) The nozzle and fittings to be easily and quickly inspected

Correct Answer: B (The nozzle end to be run out to the fire)

Explanation: Fire hoses are stowed at the hydrant so they can be run out rapidly toward the scene of a fire without kinks. This ensures that the crew can deploy the firefighting line immediately without losing time untangling or connecting sections.



- A) The fire pumps
- B) Exposed hard piping
- C) The hydrant valves
- D) The fire hose

Correct Answer: D (The fire hose)

Explanation: Fire hoses are the most vulnerable component because they are made of flexible, woven material exposed to physical damage, abrasion, heat, and potential rotting. Regular inspection is required to ensure they do not burst under fire main pressure.

Q9. In what location MUST a duplicate fire control plan be located?

- A) Gangway
- B) Engine Room
- C) Crew Mess
- D) Chief Mate's Office

Correct Answer: A (Gangway)

Explanation: Per SOLAS regulations, a duplicate of the fire control plan must be permanently stored in a weather-tight enclosure outside the deckhouse. This allows firefighters coming from the shore to quickly assess the ship's fire safety arrangements.

Q10. A vessel's fire control plan shall be

- A) Posted every 150 feet along the most continuous deck on the vessel
- B) Permanently posted for the guidance of ship's officers
- C) Given to each crew member in booklet form
- D) Translated into three languages: English, French and Spanish

Correct Answer: B (Permanently posted for the guidance of ship's officers)

Explanation: The fire control plan serves as a vital document for officers and firefighters to understand the location of fire detection, control, and containment systems. It must be permanently posted to ensure it is always available for reference during emergencies.

Q11. The muster list shows each rig hand's muster station, his duties during abandonment, basic instructions, and

- A) All emergency signals
- B) Instructions for lowering the survival craft
- C) The time each weekly drill will be held
- D) Work schedule

Correct Answer: A (All emergency signals)

Explanation: The muster list defines the emergency organization of the ship. It must clearly outline the specific muster station for each crew member, their abandonment duties, and the universal emergency signals so all personnel know how to react instantly.

Q12. Which of the following statements is true concerning fire hose stations used in the engine room of a tanker or cargo vessel?

- A) It must be marked in red letters two inches high
- B) National Standard fire hose coupling threads shall be used
- C) In heavy weather, the hose may be removed temporarily from the hydrant

Correct Answer: B (National Standard fire hose coupling threads shall be used)

Explanation: To maintain compatibility with shore-based equipment and facilitate inter-ship assistance, SOLAS requires that all hose couplings are interchangeable and conform to established national or international standards.



- A) Canister-type gas mask
- B) Fresh-air breathing apparatus
- C) Self-contained breathing apparatus
- D) Combustible gas indicator

Correct Answer: C (Self-contained breathing apparatus)

Explanation: Cargo vessels are required to carry self-contained breathing apparatus (SCBA) as part of the fireman's outfit. Unlike canister masks, SCBAs provide an independent supply of breathable air, essential for entering smoke-filled or oxygen-deficient spaces.

Q14. The international shore connection

- A) Allows hook up of firefighting water from shore facilities
- B) Satisfies pollution prevention requirements
- C) Allows emergency use of the fire main for de-ballasting
- D) Permits discharge of waste oil to shore facilities

Correct Answer: A (Allows hook up of firefighting water from shore facilities)

Explanation: The International Shore Connection (ISC) is a universal flange that allows shore-based fire services to connect their hoses to the ship's fire main. This enables shore teams to supply water to the ship if the internal pumps fail.

Q15. Automatic fire dampers in ventilation systems are operated by use of

- A) Heat or smoke detectors
- B) CO2 system pressure switches
- C) Remotely operated valves
- D) Fusible links

Correct Answer: D (Fusible links)

Explanation: Fusible links are heat-sensitive components designed to melt at a specific temperature. When they melt, the tension holding the damper open is released, allowing the fire damper to close automatically and isolate ventilation ducts to prevent fire spread.

Q16. If passengers are on board when an abandon ship drill is carried out, they should

- A) Take part
- B) Watch
- C) Go to their quarters
- D) Stay out of the way and do what they want

Correct Answer: A (Take part)

Explanation: SOLAS regulations mandate that all passengers must participate in abandon ship drills. This ensures they are familiar with their lifejacket donning procedures, the emergency signal, and the route to their designated assembly stations.

Q17. The locker or space containing the self-contained breathing apparatus must be

- A) Located in close proximity to the bridge
- B) Equipped with battery powered emergency lighting
- C) Marked "SELF-CONTAINED BREATHING APPARATUS"

Correct Answer: C (Marked "SELF-CONTAINED BREATHING APPARATUS")

Explanation: To ensure immediate accessibility, the storage locker for SCBA must be clearly marked. This allows any crew member to locate the equipment instantly in the event of an emergency, such as a fire or gas leak.



- A) CO2
- B) DCP
- C) FOAM
- D) WATER TYPE

Correct Answer: A (CO2)

Explanation: CO2 extinguishers are difficult to recharge onboard because they require specialized high-pressure pumping equipment, chilled CO2 reservoirs, and precise weighing scales. Most vessels carry spare cylinders instead of attempting to recharge them.

Q19. What is the duration of EEBD?

- A) 05 mins
- B) 8 mins
- C) 12 mins
- D) 10 mins

Correct Answer: A (05 mins)

Explanation: Emergency Escape Breathing Devices (EEBD) are designed to provide a short duration of air to allow personnel to escape from a smoke-filled space. SOLAS requires a minimum service duration of at least 10 minutes.

Q20. What action should be taken immediately after a fire has been extinguished?

- A) They are liable to spontaneous combustion
- B) They are not a hazard
- C) They are not a hazard unless ignited
- D) They generate their own oxygen when ignited and difficult to extinguish

Correct Answer: D (They generate their own oxygen when ignited and difficult to extinguish)

Explanation: Certain materials involved in fires (like some chemicals or specific fuels) can be self-oxidizing. These require specialized extinguishing methods because they do not rely on atmospheric oxygen and are notoriously difficult to suppress once ignited.

Q21. There are two disadvantages to CO2 when used as a total flooding firefighting agent. One of them is extinguishing concentration will not support human life and

- A) The lack of cooling effect on heated materials
- B) It breaks down under extreme heat to form poisonous gases
- C) It cannot be used in a dead ship situation without electrical power available to CO2 pump
- D) There is no effect on class A fire even in an enclosed space

Correct Answer: A (The lack of cooling effect on heated materials)

Explanation: CO2 is an asphyxiant that displaces oxygen, making it deadly in enclosed spaces. A major disadvantage is its lack of cooling capacity, meaning that once the CO2 gas dissipates, the hot fuel may reignite if the surfaces have not cooled down.

Q22. Foam tends to extinguish fire by

- A) Smothering the burning material
- B) Chemically absorbing oxygen
- C) Releasing large amount of CO2
- D) None of the above

Correct Answer: A (Smothering the burning material)

Explanation: Foam extinguishes fires primarily by creating a coherent blanket that 'smothers' the fire, cutting off the supply of



~~Q23. If restricted visibility is encountered in an accommodation fire, you would~~

- A) Carry a torch
- B) Stand up straight to keep your head above the smoke
- C) Use a fan to blow the smoke and flames
- D) Rig lifelines to assist a retreat

Correct Answer: D (Rig lifelines to assist a retreat)

Explanation: In smoke-filled compartments with limited visibility, lifelines are essential for safety. They guide firefighters back to the exit point, ensuring that even if they become disoriented or visibility drops to zero, they can find their way out.

Q24. Firefighters should come back from the area

- A) After 10 mins of firefighting
- B) Upon instruction from another firefighter
- C) Upon hearing the warning sound from SCBA
- D) None of the above

Correct Answer: C (Upon hearing the warning sound from SCBA)

Explanation: The SCBA's low-pressure warning alarm is a critical safety device that triggers when the air cylinder is nearing depletion. Firefighters must immediately withdraw from the hazardous zone upon hearing this signal to ensure they reach safety with a reserve of air.

Q25. In portable fire extinguishers, PASS stands for

- A) Pass, Aim, Squeeze, Sweep
- B) Pass, Aim, Shut, Sit
- C) Pass, At, Sure, Speed
- D) Pull, Aim, Squeeze, Sweep

Correct Answer: D (Pull, Aim, Squeeze, Sweep)

Explanation: PASS is the standard acronym for operating a fire extinguisher: Pull the pin, Aim the nozzle, Squeeze the lever, and Sweep the agent across the base of the fire.

Q26. The color coding of a portable extinguisher is

- A) Signal Red
- B) Blue
- C) Black

Correct Answer: A (Signal Red)

Explanation: According to international standards for firefighting equipment, the base color for portable fire extinguishers is Signal Red. This ensures they are easily identifiable against ship bulkheads.

Q27. Automatic CO2 release system is for which space?

- A) Accommodation
- B) Main Deck
- C) Engine Room
- D) None of the above

Correct Answer: C (Engine Room)

Explanation: Automatic CO2 release systems are typically installed in machinery spaces. These systems are triggered by detection systems to provide a rapid, total-flooding extinguishing response where manual intervention might be dangerous.



- A) French
- B) English
- C) Spanish
- D) Any local language

Correct Answer: B (English)

Explanation: The fire control plan must be written in the ship's working language, which is typically English, as it is the international maritime language for safety and operations. This ensures all crew members can effectively interpret the emergency information.

Q29. Class-A fire extinguisher is best used on?

- A) Solids
- B) Liquids
- C) Gas
- D) Metals

Correct Answer: A (Solids)

Explanation: Class A fires involve ordinary solid combustible materials such as wood, paper, textiles, and rubber. Water-type extinguishers are the primary choice for cooling and extinguishing these types of fires.

Q30. What is the maximum weight of a portable fire extinguisher?

- A) 16 Kg
- B) 25 Kg
- C) 12 Kg
- D) 23 Kg

Correct Answer: A (16 Kg)

Explanation: Portable fire extinguishers must be portable by a single person to ensure rapid response. Regulation limits the weight of a portable extinguisher to a maximum of 23 kg, though 16 kg is often the practical upper limit for ergonomic handling.